

Funding Discontinuities and Arbitrage-Free Pricing of Perpetual Futures

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Abstract

Kalshi’s recent perpetual futures design introduces a discontinuous funding rule with zero funding inside an inaction region and a discrete jump outside it. This note shows that such a specification can eliminate stationary arbitrage-free futures-to-spot ratios when the net cost of carry is moderate but nonzero. In this region, funding is either too low or too high to rule out arbitrage. The resulting trades and price pressure are illustrated.

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Following recent CFTC approval, Kalshi has started trading perpetual futures for US-based investors. Kalshi’s contract specifications differ somewhat from those of established offshore trading venues such as BitMEX, Binance, and Hyperliquid. Two features stand out. First, Kalshi introduces discontinuities in the funding-rate specification on either side of an inaction region with zero funding. Established venues use nonlinear transformations, such as clamps, but these do not introduce discontinuities. Second, Kalshi values funding payments based on the perpetual price, while established venues have relied mostly on the spot or oracle price. This note studies the implications of these choices for arbitrage-free prices of perpetuals.

The main finding of our analysis is that discontinuities create a region where stationary arbitrage-free futures-to-spot ratios do not exist. In the frictionless case we analyze, this implies arbitrage trades that push futures-to-spot ratios toward the boundaries of the inaction region, with funding rates switching between zero and slightly above 1 bp as the ratio moves from one side of a boundary to the other. In a reasonably frictionless market, this suggests

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that futures-to-spot ratios may spend little time between the funding inaction cutoffs at 0.9999 and 1.0001. These effects are unlikely to be large for directional traders, but they may negatively affect levered basis trading and market making, where small pricing discrepancies and predictable funding carry are central.

We first present a pricing model for perpetual futures that captures some of Kalshi’s key design choices for funding payments and derive arbitrage-free prices. We show that within the inaction region, except for the knife-edge case where net cost of carry is exactly zero, there are no stationary arbitrage-free futures-to-spot ratios. For this case, we illustrate the resulting arbitrage trades, discuss their practical relevance, and consider extensions and alternative funding specifications.

1 Perpetual futures model with continuous funding payments

The P&L for a long position in Kalshi’s perpetual futures contract in a discrete-time setup can be defined as

$$f_{t+1} - f_t - Fund_{t+1},$$

with f_t the futures price, say USD per BTC. As a starting point, funding payments are defined as

$$Fund_{t+1} = r_t^f f_{t+1}$$

and the funding rate r_t^f is given by

$$r_t^f = \frac{f_t - x_t}{x_t}.$$

Funding payments anchor futures to spot by requiring higher transfers from longs to shorts when futures prices are higher relative to spot, x_t . Exchanges typically apply some nonlinear transformation, such as clamps. To establish a reference point, we start our analysis without such transformations, and as consequence funding rates are continuous in the gap between futures and spot. Exchanges determine funding rates as averages computed at frequent intervals over the funding period. For simplicity, this analysis ignores this averaging. The funding rate is applied to the notional value, which here is measured using the futures price. This is in line with Kalshi’s specifications (KalshiEX, 2026) but not those of the more established perp trading venues.

The absence of arbitrage implies that

$$E_t^{\mathbb{Q}} \left[f_{t+1} - f_t - \left(\frac{f_t - x_t}{x_t} \right) f_{t+1} \right] = 0,$$

which can be rewritten as

$$E_t^{\mathbb{Q}} \left[\frac{x_{t+1}}{x_t} \frac{f_{t+1}}{x_{t+1}} - \frac{f_t}{x_t} - \left(\frac{f_t}{x_t} - 1 \right) \frac{x_{t+1}}{x_t} \frac{f_{t+1}}{x_{t+1}} \right] = 0. \quad (1)$$

The absence of arbitrage and assuming constant interest rates imply $E_t^{\mathbb{Q}} \frac{x_{t+1}}{x_t} = \frac{1+r}{1+r^*}$, where r is the interest rate of the numeraire, say USD, and r^* is the interest rate of the underlying cryptocurrency. For more details, see Ackerer, Hugonnier and Jermann (2025).

Consider the following notation for the stationary futures-to-spot ratio q , the interest-rate ratio and the net cost of carry h ,

$$q = \frac{f_t}{x_t} = \frac{f_{t+1}}{x_{t+1}}, \quad R = \frac{1+r}{1+r^*}, \quad 1 - \frac{1}{R} = \frac{r-r^*}{1+r} = h.$$

With these definitions and assumptions, Equation (1) for a stationary futures-to-spot ratio becomes

$$Rq - q - (q - 1) Rq = 0.$$

This is a quadratic equation in q and therefore has two solutions. First, $q = 0$ is a solution. Whether such a solution is empirically relevant is an open question. It is due to the choice of valuing funding payments with the futures price. The more standard practice of valuing funding payments with the spot or oracle price does not introduce this solution (as can easily be checked). We view this solution as degenerate rather than economically relevant. At $q = 0$, the funding rate can be large in magnitude, but the funding payment is zero because the funding notional is zero. The contract therefore loses both its price-change exposure and its funding anchor. The model contains no mechanism that moves the price away from this absorbing state. To rule it out one must impose an additional admissibility condition, such as strictly positive perpetual prices for strictly positive spot prices, or modify the funding notional so that funding does not vanish when the perpetual price collapses. Thus, throughout the economically relevant analysis below, we restrict attention to positive stationary ratios, $q > 0$, while noting that the algebraic zero root is a separate degeneracy of the futures-notional specification.

The second solution solves

$$R - 1 - (q - 1) R = 0,$$

so that

$$q = 2 - \frac{1}{R} = 1 + h. \quad (2)$$

Intuitively, the futures-to-spot ratio is increasing in the net cost of carry h . For dated futures and forwards, the cost of carry also positively affects the price of the derivative contract, for instance, in the covered interest parity equation $f_t^{(1)}/x_t = \frac{1+r}{1+r^*} = \frac{1}{1-h}$.¹

The funding rate is

$$r^f = q - 1 = h.$$

The fair-value funding rate, r^f , is equal to the net cost of carry h .

2 Discontinuous clamp-like funding specification

Consider the following funding-rate specification, which mimics Kalshi's specification:

$$r_t^f = \begin{cases} 0, & \text{for } \left| \frac{f_t - x_t}{x_t} \right| \leq \varepsilon \\ \frac{f_t - x_t}{x_t}, & \text{for } \left| \frac{f_t - x_t}{x_t} \right| > \varepsilon. \end{cases}$$

There is an inaction region, or deadband, typically $\varepsilon = 0.0001$ for an 8-hour funding period, with $0.0001 \times 365 \times 3 \approx 11\%$ per annum. The main novelty of this specification is that it introduces a discontinuity at the edges of the inaction region. The funding rate jumps from 0 to 1 bp. The standard approach so far at BitMEX, Binance, and Hyperliquid has been to use a clamp function, which also has a region of inaction but no discontinuity. Kalshi's specification is intuitively simpler, but the discontinuity may have some undesirable effects.

The no-arbitrage conditions are now

$$E_t^{\mathbb{Q}}[f_{t+1} - f_t] = 0 \quad \text{for } \left| \frac{f_t - x_t}{x_t} \right| \leq \varepsilon$$

$$E_t^{\mathbb{Q}} \left[f_{t+1} - f_t - \left(\frac{f_t - x_t}{x_t} \right) f_{t+1} \right] = 0 \quad \text{for } \left| \frac{f_t - x_t}{x_t} \right| > \varepsilon.$$

We solve for stationary equilibrium prices by computing stationary prices in each region and then verifying that they are admissible equilibrium prices.

¹A more standard oracle-based funding specification $Fund_{t+1} = \left(\frac{f_t - x_t}{x_t} \right) x_{t+1}$ implies an arbitrage-free stationary futures-to-spot ratio satisfying $Rq - q - (q-1)R = 0$, with the unique solution $q = R = 1/(1-h) \approx 1+h$.

Region 1 (inaction)

$$E_t^{\mathbb{Q}} \left[\frac{x_{t+1}}{x_t} \frac{f_{t+1}}{x_{t+1}} - \frac{f_t}{x_t} \right] = 0$$

For the constant interest-rate ratio $E_t^{\mathbb{Q}} \frac{x_{t+1}}{x_t} = \frac{1+r}{1+r^*} = R$, a stationary no-arbitrage futures ratio satisfies

$$[R - 1] q = 0,$$

and there are two solutions to this equation: $q = 0$, and if $[R - 1] = 0$, any value for q is a solution. The second is a knife-edge case.

Are these solutions equilibria? An equilibrium in this region requires

$$|q - 1| \leq \varepsilon$$

Clearly, $q = 0$ is not an equilibrium because $|q - 1| = 1 > \varepsilon = 0.0001$. For the knife-edge case with $[R - 1] = 0$, any q satisfying $-\varepsilon \leq q - 1 \leq \varepsilon$ is an equilibrium.

Region 2 (action). The algebra is the same as above. The stationary ratio q solves

$$Rq - q - (q - 1) Rq = 0$$

which has a formal zero-price solution $q = 0$ and a second solution

$$q = 2 - \frac{1}{R} = 1 + h.$$

For an equilibrium to be in this region it is required that

$$|q - 1| > \varepsilon.$$

In this case, $q = 0$ is a formal equilibrium. For the nondegenerate fixed point, substituting in the solution

$$\left| 1 - \frac{1}{R} \right| = |h| > \varepsilon$$

shows that a sufficiently large net cost of carry, positive or negative, places the futures-to-spot ratio in this region.

Summary for discontinuous clamp-like function

(1) If $|h| > \varepsilon$ (large net cost of carry), funding is in the action region and the economically relevant solution is $q = 1 + h$.

(2) If $0 < |h| \leq \varepsilon$ (moderate but nonzero net cost of carry), no stationary arbitrage-free price ratio exists.

(3) If $h = 0$ (knife-edge), funding is in the inaction region, and any q such that $-\varepsilon \leq q - 1 \leq \varepsilon$ is arbitrage-free.

Most interesting is the region for which no stationary arbitrage-free price ratio exists. What does this imply practically? The next section provides some answers.

3 For moderate net cost of carry, there is arbitrage

As shown, for moderate but nonzero cost of carry, specifically for $0 < |h| \leq \varepsilon$, arbitrage-free stationary futures-to-spot ratios do not exist. This section gives examples of arbitrage trades assuming stationary futures-to-spot ratios and shows that if $0 < |h| \leq \varepsilon$ any stationary futures-to-spot ratio implies arbitrage.

3.1 Setup and notation

With stationary futures-to-spot ratios

$$f_t = qx_t \text{ and } f_{t+1} = qx_{t+1}.$$

The funding rule written more compactly is

$$g(q) = r_t^f = \begin{cases} 0, & \text{for } |q - 1| \leq \varepsilon \\ q - 1, & \text{for } |q - 1| > \varepsilon. \end{cases}.$$

The long-side perp P&L is

$$f_{t+1} - f_t - g(q) f_{t+1}$$

and with stationarity

$$f_{t+1} - f_t - g(q) f_{t+1} = (1 - g(q)) qx_{t+1} - qx_t.$$

3.2 Hedge 1: long perp, short spot

Consider the strategy of being long one perp and short

$$(1 - g(q)) \frac{q}{1 + r^*}$$

units of the underlying asset. Shorting spot means borrowing the underlying, selling immediately, and buying it back at $t + 1$ to extinguish the debt in the underlying (with appropriate

interest). At time t , shorting this spot amount gives cash proceeds in units of the numeraire (dollars)

$$(1 - g(q)) \frac{q}{1 + r^*} x_t.$$

The proceeds are invested at the numeraire interest rate so that this will be worth

$$(1 - g(q)) \frac{q}{1 + r^*} x_t (1 + r)$$

at $t + 1$. Now buy to deliver, including underlying interest,

$$(1 - g(q)) \frac{q}{1 + r^*} (1 + r^*) = (1 - g(q)) q$$

at cost

$$(1 - g(q)) q x_{t+1}.$$

Therefore, the short-spot/cash leg pays

$$(1 - g(q)) q x_t R - (1 - g(q)) q x_{t+1}.$$

Now add the long-perp payoff

$$(1 - g(q)) q x_{t+1} - q x_t + (1 - g(q)) q x_t R - (1 - g(q)) q x_{t+1}$$

which simplifies to

$$q x_t [(1 - g(q)) R - 1].$$

Therefore, since $q x_t > 0$, whenever

$$(1 - g(q)) R > 1$$

equivalently

$$g(q) < h$$

the strategy long perp, short spot is an arbitrage. The intuition is that whenever the funding cost is lower than the net cost of carry, one can go long futures and short spot for a profit.

3.3 Hedge 2: short perp, long spot

Mutatis mutandis, if

$$g(q) > h$$

the strategy short perp, long spot is an arbitrage.

Thus, the whole question reduces to comparing the funding rate $g(q)$ to the net cost of carry h . Depending on which is bigger, for $qx_t > 0$, either $qx_t [(1 - g(q))R - 1]$ or $-qx_t [(1 - g(q))R - 1]$ is positive and represents a sure positive payoff with zero initial net investment. This is arbitrage. Can $g(q)$ and h ever be equal for moderate cost of carry, $0 < |h| \leq \varepsilon$? No, because funding rates jump discretely between 0 and ε , so the funding rate is either too small or too large to rule out arbitrage.

4 Discussion

Consider a numerical example. The inaction range is typically set at $\varepsilon = 0.0001$, and a typical net cost of carry for an 8-hour period $h = \frac{r-r^*}{1+r} = 0.00005$ which can represent a USD rate of about 5% per annum and a crypto rate of $r^* = 0$. This puts us squarely into the moderate cost-of-carry region for which there is no stationary arbitrage-free futures-to-spot ratio. Suppose the market somehow sets the futures-to-spot ratios at

$$q = \frac{f}{x} = 1$$

which implies a funding rate of

$$r^f = g(q) = 0.$$

In this case, we have

$$g(q) = 0 < h = 0.00005$$

and there would be buying pressure in the perps market because zero funding payments are a good deal for the longs. Alternatively, suppose the market arrives at a higher price level, say slightly above $q = 1.0001$, where the funding rate has switched from 0 to just above 0.0001. Now, the funding rates are higher than the net cost of carry and arbitrageurs go short, and there would be selling pressure on perps. With the price being close to $q = 1.0001$, there is no resting point because funding payments are either at 0 or just above 1 bp, which is either too low or too high to stop arbitrage trading. To support an arbitrage-free market, the funding rate should be somewhere in between 0 and 1 bp, but the discontinuity in the funding rate specification does not allow it.

If interest rates were different and the net cost of carry were moderate and negative, say $h = \frac{r-r^*}{1+r} = -0.00005$, futures-to-spot ratios would hover around $q = 0.9999$.

Extrapolating beyond the stationary model, one may conjecture that futures prices spend little time inside the inaction band between 0.9999 and 1.0001 and instead bunch near the

relevant cutoff.

The inaction range is not economically negligible. For $\varepsilon = 0.0001$, a stationary arbitrage-free futures-to-spot ratio outside the inaction region requires a net cost of carry satisfying $|h = \frac{r-r^*}{1+r}| > 0.0001$, which in annualized terms is $0.0001 \times 3 \times 365 = 10.95\%$.

How economically important are these price and funding effects? Directional traders are unlikely to be affected much. However, for levered basis trading and market making, small pricing discrepancies and predictable funding are important. By making funding carry less smooth near the inaction boundaries, the discontinuity may reduce the arbitrage capital willing to keep futures prices tightly aligned with spot.

Alternatives are available. The standard clamp does not have this discontinuity. However, as shown in Hugonnier and Jermann (2025), there is a tendency for probability mass to accumulate around the clamp inaction points, which are typically around $f_t/x_t \in \{0.99995, 1.00005\}$. Thus, the standard clamp alleviates this issue but does not fully resolve it.

As another alternative, the funding payments could be unfiltered by any clamp or clamp-like transformation. For perp markets that are sufficiently deep, this might be a better choice. See also the quantitative analysis in Hugonnier and Jermann (2025).

Nonstationary futures-to-spot ratios are mathematically possible in the model, but they do not resolve the problem in an economically attractive way. If the funding rule cannot deliver the small nonzero funding rate required by the cost of carry, then a fair price must instead be supported by expected changes in the futures-to-spot ratio. Under simple deterministic dynamics, these paths are unstable rather than stabilizing. Depending on the initial ratio and the sign of the net cost of carry, the ratio either moves toward the degenerate zero-price root or moves upward toward a singularity which is followed by negative values. See the appendix for more details.

All explicit solutions in this note were derived under constant interest rates. A natural extension is to allow stochastic interest rates. In principle, stochastic continuation values can support stationary arbitrage-free prices even when some states lie inside the funding-rate inaction region. In the empirically plausible calibrations we considered, however, this did not resolve the nonexistence problem. The reason is intuitive. The purpose of an inaction region is that futures-to-spot ratios close to one should occur with nontrivial frequency. But these are precisely the states in which the required funding adjustment is likely to be small but nonzero, while the discontinuous rule sets funding to zero. In our numerical experiments, calibrations that placed more than a very small number of states inside the inaction region typically failed to admit a stationary arbitrage-free price vector.

The analysis is deliberately frictionless. It abstracts from bid-ask spreads, execution costs, margin constraints, liquidation risk, price impact, and the averaging of funding rates over

the funding interval. These frictions may limit the size of the arbitrage trades in practice, but they do not eliminate the fact that discontinuities in the funding rate generate gaps in attainable funding rates, which can interfere with the smooth operation of the market.

References

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KalshiEX, CFTC Regulation 40.2(a) Notification Regarding the Initial Listing of the BTCPERP Futures Contract, June 2, 2026

Appendix: Nonstationary dynamics

The no-arbitrage relation is

$$E_t^{\mathbb{Q}} [(1 - g(q_t)) f_{t+1} - f_t] = 0$$

scaled by x_t

$$E_t^{\mathbb{Q}} \left[(1 - g(q_t)) \frac{x_{t+1}}{x_t} q_{t+1} - q_t \right] = 0.$$

Through a change of measure from numeraire to the underlying, the no-arbitrage equation can also be expressed as

$$(1 - g(q_t)) R_{t,t+1} E_t^{\mathbb{Q}^*} [q_{t+1}] - q_t = 0.$$

This representation removes the direct stochastic exchange-rate term from the pricing equation and shows that the current futures-to-spot ratio is pinned down by the appropriately risk-adjusted expected future ratio.

The change-of-measure representation does not rule out stochastic paths for q_t , even with constant interest rates. However, with constant interest rates and no additional state variables, such paths would have to be driven by shocks that are not otherwise specified in the model. They are therefore naturally interpreted as extraneous sunspot-like components rather than fundamental arbitrage-free prices. As in standard forward-looking rational-expectations models, an admissibility or no-bubble restriction selects the solution tied to the modeled fundamentals. Under this restriction, stochastic nonstationary paths do not provide an economically meaningful substitute for a stationary futures-to-spot ratio.

Under constant interest rates and deterministic q_{t+1}

$$(1 - g(q_t)) R q_{t+1} = q_t$$

and

$$q_{t+1} = \frac{q_t}{R(1 - g(q_t))} = \frac{q_t(1 - h)}{1 - g(q_t)}.$$

With the discontinuous funding rule this defines a deterministic recursion

$$\begin{aligned} q_{t+1} &= q_t(1 - h) \quad \text{if } |q_t - 1| \leq \varepsilon \\ q_{t+1} &= \frac{q_t(1 - h)}{2 - q_t} \quad \text{if } |q_t - 1| > \varepsilon. \end{aligned} \tag{3}$$

Now consider the case with moderate cost of carry $0 < |h| \leq \varepsilon$.

If $h > 0$

Inside the inaction region, $q_{t+1} < q_t$, and once below the inaction region q falls towards 0.

If starting above the upper boundary, with $q_t > 1 + \varepsilon$, close to the boundary this is increasing

$$q_{t+1} = \frac{q_t(1-h)}{2-q_t} > q_t$$

and q_{t+1} will eventually breach 2 and the next q_{t+1} is negative.

For $h < 0$, the directions are reversed: inside the inaction region the ratio moves upward; once above the upper boundary it moves toward the singularity at $q_t = 2$, while paths starting below the lower boundary move toward zero.

The dynamic system defined by the no-arbitrage relation, Equation (3), has no independent physical state variable once interest rates are constant. The futures-to-spot ratio is therefore a forward-looking, or jump, variable. As in standard rational-expectations models, admissibility conditions can be used to select among formal paths. Here, positive perpetual prices rule out paths that move toward the degenerate zero-price root or leave the positive-price region after reaching the singularity at ($q_t = 2$). Thus nonstationary paths do not provide an economically admissible substitute for a stationary arbitrage-free ratio. When a nondegenerate stationary solution exists, it is the natural admissible solution; in the moderate-cost-of-carry case considered here, no such solution exists.